SUPPORTING ADULT GUIDANCE

Estimating and judging reasonableness

Number Systems and Calculator Judgement · Lesson 7. For any supporting adult — teacher, practitioner, parent or carer. **You do not need to be a mathematics specialist to support this lesson.**

What this lesson does

This final lesson of the unit brings estimation to Higher level: rounding to 1 significant figure with powers, roots, decimals less than 1 and fractions; locating square roots to 1 decimal place; and deciding whether an estimate is an overestimate or an underestimate. In the main activity, learners round, estimate, judge someone's answer and decide the direction, building a decision trail. They then trap a true answer between a deliberate overestimate and underestimate.

Driving Question: *Is my answer about right?*

The mathematics, in plain terms

For positive numbers, increasing a factor or a numerator increases a quotient or product; increasing a divisor decreases it. So if every rounding pushes the result the same way, the estimate is a guaranteed overestimate or underestimate; if they push different ways, its direction cannot be known without more work. Choosing roundings deliberately gives two bounds that trap the true value.

The worked method the learner sees

These are the two worked examples the learner meets on screen, with the lesson's exact method and notation. They are here for consistency, not because the mathematics is demanding — whether you teach mathematics or are meeting it afresh alongside the learner, working from the same steps and the same language keeps your support aligned with what is on the screen, and lets you point back to a line the learner has already seen. Skip them freely if the method is already familiar.

Example 1 — The sense detector

Someone says 812 ÷ 3.9 = 20.82. Is that sensible?

1. Round each number to 1 significant figure.	812 → 800, 3.9 → 4
2. Estimate.	800 ÷ 4 = 200
3. Compare: 20.82 is about 10 times smaller than 200.	not sensible
4. Direction: 812 rounded down ↓; 3.9 rounded up, and we divide by it ↓.	an underestimate

20.82 is not sensible: the answer is just over 200 (it is 208.2…).

Example 2 — Locating a root

Estimate √50 to 1 decimal place.

1. Find the squares either side.	$7^2 = 49$, $8^2 = 64$
2. 50 is much nearer 49.	√50 is just over 7
3. Test a guess by squaring it.	$7.1^2 = 50.41$
4. 50 is between $7.0^2 = 49$ and $7.1^2 = 50.41$, and much nearer 50.41.	√50 ≈ 7.1

√50 ≈ 7.1 to 1 decimal place.

Judge with reasons

Learners first secure 1 significant figure (including the tricky small decimals), then locate roots by squaring, then use both in the sense detector, where every decision is recorded with its reason. The trap activity turns the direction idea into a creative task. A stretch proves why numerator up / denominator down must overestimate. A calculator may be used to check the true answer at the end — but always after the estimate.

The shape of the lesson

Timings are invitations, not deadlines. A learner may need much longer on one phase and skim another — that is expected. Each Cornerstone is given an honest mathematical role here:

Cornerstone	Its role here	Invitational timing
Connection	One significant figure: eight numbers, including 0.0372 and 1495.	~6 min
Movement	The root locator: squares either side, nearer which, then square a 1 d.p. guess.	~9 min
Reflection	The sense detector: round, estimate, judge, decide the direction; a decision trail. Stretch: prove the overestimate rule.	~16 min
Creativity	Trap the answer: a deliberate overestimate and underestimate.	~9 min
Rest	An estimate as a glance at a map.	~2 min
Nutrition	Intellectual: number sense, and trusting your numbers.	~3 min

Common misconceptions, and gentle repairs

These are the sticking points to expect. In every case, the aim is a question that returns the thinking to the learner — never to supply the answer.

Rounding small decimals to 0 or to the wrong place (0.0372 → 0.4 or 0).
Ask: “Which is the first digit that isn’t 0? Which place is it in?”

Thinking dividing always makes smaller (800 ÷ 0.4 = 320).
Ask: “How many 0.4s are there in 1? In 800?”

Halving to find a square root (√50 ≈ 25).
Ask: “What is 25^2?”

Believing rounding everything up always gives an overestimate, even for a divisor.
Ask: “If you divide by a bigger number, does the answer get bigger or smaller?”

Trusting the calculator display without comparing.
Ask: “About how big should it be?”

Protecting the support pathway

The lesson offers support in a deliberate order: **Socratic prompt** → **first try** → **hint** → **second try** → **worked solution**. The worked solution for each question stays locked until the learner has used the prompt, used the hint, and checked two answers. This is by design: the steps *are* the learning.

Please hold this line
Do not read the worked solution aloud, or talk a learner straight to the answer, while it is still locked. If they are stuck, use the Reconnection Routes, the prompt and the hint, or ask one of the repair questions above. Arriving at a correct line of reasoning — even slowly — is worth far more than being handed the result.

Reconnection Routes

If an earlier idea is blocking progress, the lesson’s Reconnection Routes offer short recaps of exactly the prerequisites this lesson needs: rounding, square numbers, and estimating before you calculate. A learner can choose one independently, or you can invite one without requiring it. Using a route is part of learning, not evidence of falling behind, and it always returns the learner to their place.

Supporting through questions: an example

Suppose a learner says 30 × 8 ÷ 3 is an overestimate of 29.4 × 7.83 ÷ 2.6, because “I rounded everything up”. The aim is a question that lets them find the flaw themselves.

- “Which numbers are multiplied, and which do you divide by?”
- “If you divide by 3 instead of 2.6, is the answer bigger or smaller?”

- “So which way does each rounding push the estimate?”
- “Can you be sure which way it leans?”

If they stall, that is the moment for a Reconnection Route or the lesson’s prompt and hint — not for the answer.

Questions you might have

Can learners use calculators?
Yes, to check at the end — but the estimate always comes first, as in Lesson 3.

Why round under a √ to a square number rather than 1 significant figure?
Because the point is an easy root: √49 is easy, √50 is not. This is a sensible estimation choice, not a rule about significant figures.

Is the decision trail recorded anywhere?
Only on the learner's screen, for them to use in the Final Task. Nothing is scored.

What comes next?
This completes the unit. An end-of-unit check is planned; the next unit is Algebraic Expressions and Identities.

Keeping things regulated and calm

- Keep to one clear step at a time; there is no time pressure and no benefit to speed.
- Treat a wrong or unfinished answer as information for the next step, never as failure.
- Let the learner choose how to record — typed, written, drawn, spoken or annotated.
- If frustration rises, it is fine to pause and return later; the lesson holds their place.
- Avoid any language about age, school year or pace as a judgement of ability.

Recognising progress

Progress is described as Approaching Proficiency → Developing Proficiency → Proficiency → Refining Proficiency across understanding, application, communication and reflection. These describe where a learner is right now, not labels of what they are. Real understanding shown in part is still real understanding, even while other pieces are still settling.

Where they are	What it can look like
Approaching Proficiency	Beginning to engage; names some features and makes a start with support.
Developing Proficiency	Carries out the method with prompts; can explain part of the reasoning.
Proficiency	Works through independently and explains why each step is true.

Refining Proficiency	Applies the idea confidently to a new or unfamiliar case, and could teach it to someone else.

A note on evidence

Evidence can be captured in whatever form suits the learner — a photo of working, a short spoken explanation, an annotated Scratchpad image, or a written solution. There is no single required format. Incomplete work is information for the next connection, never a failure: it tells you exactly where to begin next time. The aim is a true record of the learner's thinking, not a tidy performance.

If a learner is stuck or distressed

Slow the pace and reduce the load: return to one concrete, familiar case and a single small question. Offer a Reconnection Route. Remind them they can pause and come back. A short, successful, low-stakes step rebuilds confidence more reliably than pushing through. Connection before curriculum — always.

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